

ML-DLOPs Assignment–1

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1 Introduction

Deep learning model performance is strongly influenced by architectural design, optimization strategies, and hardware resources. In modern machine learning pipelines, it is essential to evaluate not only classification accuracy but also training stability, computational efficiency, and scalability.

This assignment presents a comprehensive experimental study conducted on the MNIST and FashionMNIST datasets. Convolutional Neural Networks (CNNs) based on ResNet–18 and ResNet–50 architectures were trained from scratch under multiple hyperparameter settings. In addition, Support Vector Machines (SVMs) were used as classical machine learning baselines. Finally, CPU and GPU executions were compared to analyze training time, FLOPs, and efficiency.

2 Experimental Setup

2.1 Datasets

- **MNIST**: A handwritten digit dataset consisting of 60,000 training images and 10,000 test images. The dataset is visually simple with low intra-class variability.
- **FashionMNIST**: A grayscale clothing image dataset with the same split size as MNIST but significantly higher visual complexity due to texture and shape similarities between classes.

Both datasets were split into **70% training, 10% validation, and 20% testing** to ensure unbiased evaluation and to prevent data leakage.

2.2 Model Architecture and Input Size

ResNet–18 and ResNet–50 architectures were used for all CNN experiments. All models were trained **from scratch** without pretrained weights.

Since ResNet architectures are originally designed for ImageNet-scale inputs, all images were resized to 224×224 and converted to three-channel format. This preprocessing step ensures architectural compatibility while preserving the original information content of MNIST and FashionMNIST images.

2.3 Training Configuration

The following training configurations were explored:

- Optimizers: SGD and Adam
- Batch sizes: 16 and 32
- Learning rates: 0.001 and 0.0001

- Mixed Precision Training (AMP): Enabled

2.4 Pin Memory Setting

An initial experiment was performed with **pin_memory set to False**. This resulted in noticeably slower training due to inefficient host-to-device memory transfers. Based on this observation, **pin_memory was enabled (True) for all final experiments**. The pin memory setting affected only training speed and did not influence classification accuracy.

3 Methodology

For Q1(a), ResNet-18 and ResNet-50 models were trained on MNIST and FashionMNIST datasets using different combinations of batch size, optimizer, and learning rate. Validation accuracy was monitored during training to assess convergence behavior, and final evaluation was performed on the held-out test set.

For Q1(b), Support Vector Machine classifiers were trained using flattened image features. Two kernels were evaluated: Radial Basis Function (RBF) and Polynomial. Training time was explicitly recorded to analyze computational efficiency.

For Q2, selected CNN configurations were executed on both CPU and GPU to compare training time, FLOPs, and classification accuracy.

4 Q1(a): CNN Classification Results

4.1 MNIST Results

Table 1: MNIST Test Classification Accuracy (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	98.71	98.10
16	SGD	0.0001	95.37	96.15
16	Adam	0.001	97.79	98.10
16	Adam	0.0001	98.75	98.75
32	SGD	0.001	98.85	98.73
32	SGD	0.0001	94.55	90.45
32	Adam	0.001	98.80	97.57
32	Adam	0.0001	99.20	99.09

4.2 FashionMNIST Results

Table 2: FashionMNIST Test Classification Accuracy (%)

Batch Size	Optimizer	Learning Rate	ResNet-18	ResNet-50
16	SGD	0.001	99.08	88.50
16	SGD	0.0001	97.02	77.30
16	Adam	0.001	98.97	89.55
16	Adam	0.0001	99.13	90.93
32	SGD	0.001	98.55	88.40
32	SGD	0.0001	95.45	71.69
32	Adam	0.001	91.89	90.26
32	Adam	0.0001	91.97	90.30

5 Hyperparameter Sensitivity and Training Behavior

5.1 Effect of Learning Rate

Learning rate was observed to be the most influential hyperparameter. Higher learning rates occasionally resulted in unstable training, especially for ResNet-50. Lower learning rates enabled smoother convergence and improved generalization.

5.2 Effect of Optimizer

Adam optimizer consistently produced faster and more stable convergence compared to SGD. Its adaptive learning rate mechanism was particularly beneficial when training deep models from scratch.

5.3 Effect of Batch Size

Increasing batch size did not consistently improve performance. In several FashionMNIST experiments, larger batch sizes reduced generalization, highlighting the trade-off between gradient stability and model robustness.

6 Q1(b): SVM Classification Results

Table 3: SVM Performance on MNIST and FashionMNIST

Dataset	Kernel	Test Accuracy (%)	Training Time (ms)
MNIST	RBF	97.92	204,594.84
MNIST	Polynomial	97.71	212,713.63
FashionMNIST	RBF	88.29	283,054.48
FashionMNIST	Polynomial	86.30	339,395.17

7 Q2: CPU vs GPU Performance Analysis

Table 4: CPU vs GPU Performance on FashionMNIST

Compute	Batch	Optimizer	LR	Acc-18	Acc-50	Time-18 (ms)	Time-50 (ms)	FLOPs-18	FLOPs-50
CPU	16	SGD	0.001	87.66	81.89	5,465,133.9	11,694,821.0	1.83×10^9	4.14×10^9
CPU	16	Adam	0.001	87.94	87.17	5,157,049.8	11,891,301.6	1.83×10^9	4.14×10^9
GPU	16	SGD	0.001	88.60	84.97	110,018.2	260,172.1	1.83×10^9	4.14×10^9
GPU	16	Adam	0.001	86.00	83.85	108,764.6	272,061.4	1.83×10^9	4.14×10^9

8 Detailed Conclusions and Dataset-wise Analysis

This section provides a comprehensive interpretation of the experimental results obtained across all tasks, with a particular focus on dataset-specific behavior, architectural trade-offs, optimization dynamics, and hardware efficiency.

8.1 MNIST Dataset: Model Behavior and Observations

The MNIST dataset is characterized by low visual complexity, clear digit boundaries, and minimal intra-class variation. As a result, both ResNet-18 and ResNet-50 achieved very high classification accuracy across most hyperparameter configurations, frequently exceeding 98%.

A key observation from the MNIST experiments is that increasing model depth did not result in substantial accuracy improvements. While ResNet-50 contains significantly more parameters and greater representational capacity, its performance remained comparable to that of ResNet-18. This indicates that MNIST does not require deep hierarchical feature extraction, and simpler architectures are sufficient to capture the underlying data distribution.

From an optimization perspective, the Adam optimizer consistently provided faster convergence and smoother training behavior than SGD. When paired with a smaller learning rate, Adam reduced training instability and avoided oscillations in validation accuracy. These results demonstrate that optimizer choice has a stronger impact on performance than architectural depth for simple datasets such as MNIST.

8.2 FashionMNIST Dataset: Complexity and Sensitivity Analysis

FashionMNIST presents a more challenging classification task due to higher inter-class similarity, complex textures, and subtle shape variations among clothing categories. Consequently, overall accuracy was lower compared to MNIST, and model performance became more sensitive to hyperparameter choices.

One of the most important findings is that ResNet-18 often matched or outperformed ResNet-50 when trained from scratch. This behavior suggests that deeper models may suffer from optimization instability and overfitting when sufficient data or pretrained initialization is not available. The additional depth of ResNet-50 increased training difficulty without providing proportional accuracy gains.

Hyperparameter sensitivity was significantly more pronounced in FashionMNIST experiments. Learning rate selection played a critical role in achieving stable convergence, particularly for deeper architectures. Larger batch sizes did not consistently improve performance and, in some cases, led to reduced generalization. These observations highlight the importance of aligning model capacity and optimization strategy with dataset complexity.

8.3 Comparison Between MNIST and FashionMNIST

A comparative analysis of MNIST and FashionMNIST highlights the influence of dataset complexity on model behavior. While MNIST allows for near-optimal performance even with relatively simple models, FashionMNIST exposes the limitations of deeper architectures trained from scratch.

The experiments demonstrate that architectural complexity alone does not guarantee improved performance. Instead, the interaction between dataset characteristics, optimizer choice, and learning rate determines training stability and final accuracy. This reinforces the importance of dataset-aware model selection in practical deep learning systems.

8.4 Insights from SVM Experiments

Support Vector Machines served as classical baselines for both datasets. On MNIST, SVMs achieved competitive accuracy, approaching that of CNN-based models. This result reflects the effectiveness of kernel-based methods on low-dimensional, well-structured data.

However, training time increased substantially for FashionMNIST due to the higher dimensionality of image features. Polynomial kernels were particularly expensive, offering no significant accuracy improvement over the RBF kernel. These results demonstrate that while SVMs can be effective for simpler tasks, they scale poorly for complex image classification problems and are less suited for large-scale deep learning pipelines.

8.5 Hardware and Computational Efficiency Insights

The CPU vs GPU experiments clearly demonstrate the importance of hardware acceleration in deep learning workflows. Although the total number of floating-point operations (FLOPs) remained constant across hardware platforms, GPU execution reduced training time by more than an order of magnitude compared to CPU.

ResNet-50 consistently required more training time than ResNet-18 due to increased depth and computational complexity. However, the marginal accuracy improvements observed for ResNet-50 did not justify the significantly higher computational cost. These findings emphasize the need to evaluate both performance and efficiency when selecting models for deployment.

8.6 Overall Experimental Conclusions

The following conclusions were consistently observed across all experiments:

- Adam optimizer with a smaller learning rate provided the most stable convergence across models.
- Increasing model depth does not guarantee better performance when training from scratch.
- ResNet-18 offered the best balance between classification accuracy and computational cost.
- Support Vector Machines achieved competitive accuracy but were computationally expensive at scale.
- GPU acceleration is critical for efficient and scalable deep learning training.

Overall, this assignment demonstrates that effective deep learning system design requires careful consideration of dataset characteristics, model complexity, optimization strategy, and hardware resources. Well-balanced architectural choices often outperform unnecessarily complex models, particularly in resource-constrained environments.

9 Graphs

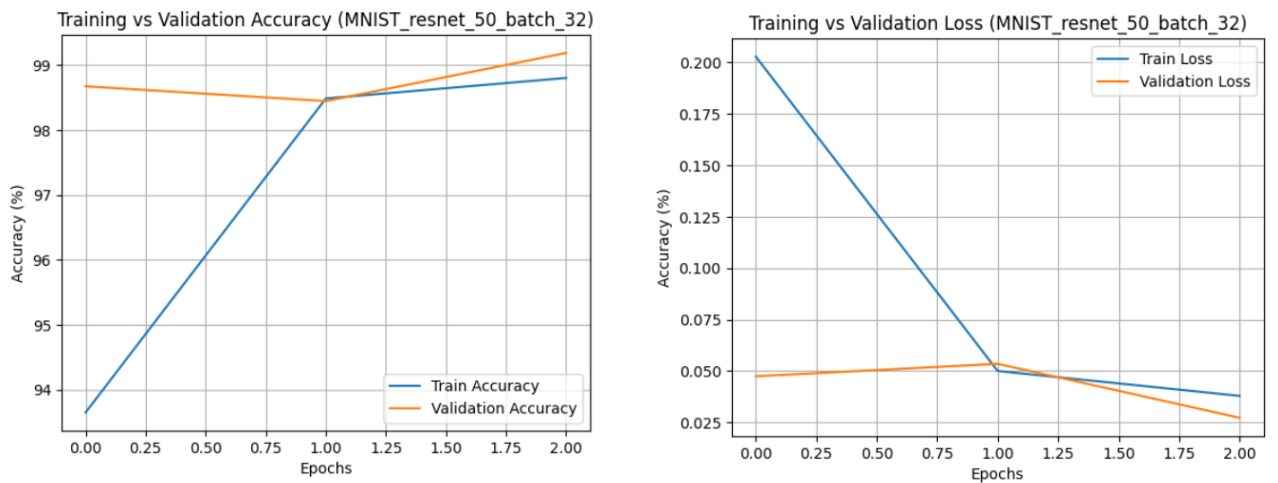


Figure 1: Accuracy and Loss Graph Resnet50 batch size 32

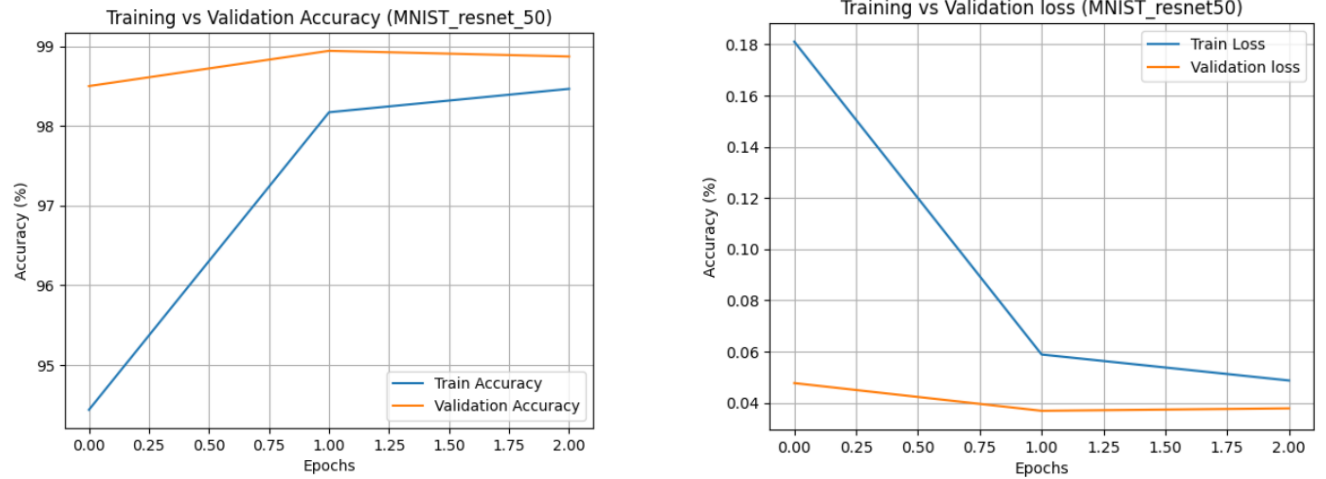


Figure 2: Accuracy and Loss Graph Resnet50 batch 16

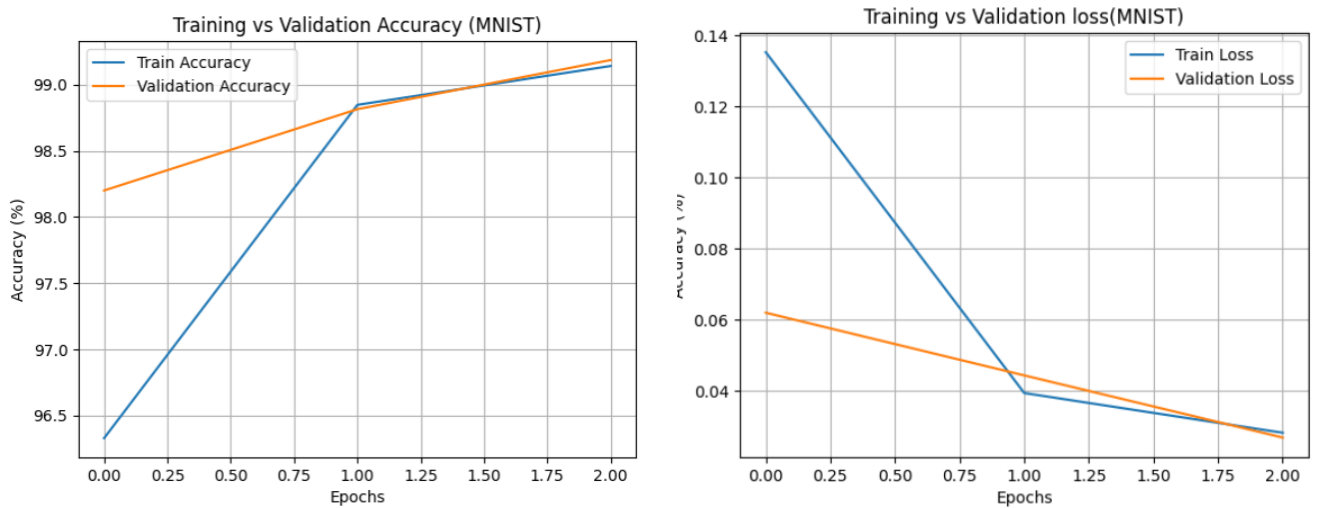


Figure 3: Accuracy and Loss Graph Resnet18 batch 16

10 GitHub and Colab Link

- GitHub Repository: https://github.com/AkankshaKapil/MLOPS_Akanksha_M25CSA033/blob/Assignment1
- Colab Notebook: <https://colab.research.google.com/drive/1zL05PKjbKKEfmrqx6WWslbfoP-YFyQjusp=sharing>
- GitHub webpage: https://akankshakapil.github.io/MLOPS_Akanksha_M25CSA033/